Lab 3.2 DMAIC report and SPC analysis

IES | 25 minutes | LO1, LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: process-data.csv, defects.csv, dmaic-template.md. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Open process-data.csv in Excel and save as a workbook. Baseline samples are 1-30 (rows 2-31); samples 31-35 are monitoring data. Fill volume is in litres. Confirm all 35 records are present and numeric.

2. Create column F called MR. Leave F2 blank; in F3 enter =ABS(B3-B2) and fill through F31 only for the baseline estimate. In H2 enter =AVERAGE(B2:B31); H3 =AVERAGE(F3:F31); H4 =H3/1.128; H5 =H2-3*H4; H6 =H2+3*H4.

3. Create an individuals line chart in sample order for all 35 values. Add constant series for the baseline mean, LCL and UCL from H2, H5 and H6. Do not recompute limits using the monitoring signal. Label litres, sample order and baseline window.

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4. Calculate teaching Cp in H7 =(10.6-9.4)/(6*H4) and Cpk in H8 =MIN(10.6-H2,H2-9.4)/(3*H4). Treat these as provisional estimates: the small baseline, stability, independence, distribution and measurement system need review. A Cp value alone cannot establish stability.

5. Open defects.csv. Sort counts descending and compute cumulative share using =SUM($B$2:B2)/SUM($B$2:$B$5), filling down. Ask Copilot for a Pareto interpretation, fishbone hypotheses and a 5-Whys chain with evidence requests, not asserted causes.

6. Use dmaic-template.md to assemble a concise report in Word, then a management-summary slide in PowerPoint. Record the monitoring signal, a proposed trial and a control plan. Compare calculations with expected-output.md before accepting the AI analysis.

Check before moving on

Baseline n=30, monitoring n=5; sample 33 signals above UCL.

Cp and Cpk use the stated within-process estimate and the 9.4-10.6 L teaching limits.

Pareto top two categories total 66.7%; causation is unproven.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 3.2 prompts and evidence

Main prompt

Use baseline samples 1-30 only to estimate an individuals chart with MRbar/1.128. Keep monitoring samples 31-35 separate. Show formulas, units, sample counts, missing-value checks and baseline limits. Flag signals without inventing a cause. Summarise defects using Pareto; label fishbone and 5-Whys statements as hypotheses. Draft all five DMAIC sections, a trial and control plan. Capability estimates are provisional, not certification.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. Baseline mean 10.01333 L; MRbar 0.18621 L; sigma estimate 0.16508 L; LCL 9.51810 L; UCL 10.50856 L; Cp 1.212; Cpk 1.185. Sample 33 is 10.8 L, above UCL. Top two defect categories are 20/30 = 66.7%. Small training sample only.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.